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TREND-BASED MOCK PAPER

GATE Electrical Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electrical Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose synonym of 'pragmatic':

(Numerical Answer Type — Enter the numerical value)

Q2. [1-MCQ] [1 mark]

Choose opposite of 'latent':

- (A) Obvious
- (B) Hidden
- (C) Clear
- (D) Manifest

Q3. [1-MCQ] [1 mark]

Distance 200km at 50km/h takes:

- (A) 2h
- (B) 4h
- (C) 5h
- (D) 3h

Q4. [1-MCQ] [1 mark]

Speed = Distance /:

- (A) Time
 - (B) Velocity
 - (C) Acceleration
 - (D) Force
-

Section B: Electrical Engineering

Q5. [1-MCQ] [1 mark]

Gain margin:

- (A) At phase crossover
- (B) At gain crossover
- (C) Both
- (D) None

Q6. [1-MCQ] [1 mark]

Root locus plots:

- (A) Open loop poles
- (B) Closed loop poles
- (C) Both
- (D) None

Q7. [1-MCQ] [1 mark]

Integral controller eliminates:

- (A) Rise time
- (B) Steady state error
- (C) Damping
- (D) Overshoot

Q8. [1-MCQ] [1 mark]

Time constant of RC:

- (A) RC
- (B) R/C
- (C) C/R
- (D) 1/RC

Q9. [2-MCQ] [2 marks]

Type 1 system with $K_v=5$. Steady-state error for unit ramp:

- (A) 0
- (B) 0.2
- (C) Infinity
- (D) 1

Q10. [2-NAT] [2 marks]

A system with characteristic $1+s = 0$ has:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-MCQ] [2 marks]

Root locus branches start at poles and end at:

- (A) Poles only
- (B) Zeros only
- (C) Zeros or infinity
- (D) Infinity only

Q12. [2-MCQ] [2 marks]

$s^2 + 2s + 5 = 0$. Damping ratio:

- (A) 0.2

- (B) 0.447
- (C) 0.707
- (D) 1.0

Q13. [2-MCQ] [2 marks]

Nyquist criterion: $N = Z - P$. If $P=1$, $N=-1$:

- (A) 1 zero in RHP
- (B) Stable
- (C) 1 clockwise encirclement needed
- (D) Unstable

Q14. [1-MCQ] [1 mark]

Step-down chopper output:

- (A) Less than input
- (B) More than input
- (C) Equal
- (D) Zero

Q15. [1-NAT] [1 mark]

Inverter converts:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-MCQ] [1 mark]

Thyristor is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q17. [1-MCQ] [1 mark]

Full-wave rectifier uses:

- (A) 1
- (B) 2
- (C) 4
- (D) 3

Q18. [2-NAT] [2 marks]

IGBT combines features of:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MSQ] [2 marks]

For a buck converter, $V_{out} =$

- (A) $D \cdot V_{in}$
- (B) V_{in}/D
- (C) $V_{in}/(1-D)$
- (D) $V_{in} \cdot D/(1-D)$

Q20. [2-MCQ] [2 marks]

Full-wave rectifier PIV across each diode:

- (A) V_m
- (B) $2 \cdot V_m$

- (C) $V_m/2$
- (D) $\sqrt{2} \cdot V_m$

Q21. [2-NAT] [2 marks]

CSI input is:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

3-phase inverter 120-degree conduction eliminates:

- (A) 3rd harmonic
- (B) 5th harmonic
- (C) 7th harmonic
- (D) No triplen

Q23. [2-MCQ] [2 marks]

Y-parameter Y₁₁ for series impedance Z is:

- (A) $1/Z$
- (B) Z
- (C) $-Z$
- (D) 0

Q24. [2-MCQ] [2 marks]

Thevenin equivalent with $V_{th}=10V$, $R_{th}=5\Omega$, load= 5Ω . Output:

- (A) 5V, 1A
- (B) 5V, 0.5A
- (C) 2.5V, 0.5A
- (D) 10V, 1A

Q25. [2-MCQ] [2 marks]

Wheatstone bridge null condition:

- (A) $R_1/R_2 = R_3/R_4$
- (B) $R_1 \cdot R_4 = R_2 \cdot R_3$
- (C) Both A and B
- (D) $R_1 + R_2 = R_3 + R_4$

Q26. [2-NAT] [2 marks]

$H(s) = (s^2 + 4s + 3)/(s^2 + 2s + 2)$. Zeros at:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Delta-Star transformation: R_a (star) =

- (A) $R_1 \cdot R_2 / (R_1 + R_2 + R_3)$
- (B) $(R_1 \cdot R_2 + R_2 \cdot R_3 + R_3 \cdot R_1) / R_3$
- (C) $R_1 + R_2$
- (D) $R_1 \cdot R_2 / R_3$

Q28. [2-NAT] [2 marks]

Two-port h-parameter h₁₁ has units:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Time constant of RL ($R=10\text{ohm}$, $L=5\text{H}$):

- (A) 2s
- (B) 0.5s
- (C) 50s
- (D) 0.05s

Q30. [1-NAT] [1 mark]

Eigenvalues of identity:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q33. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q34. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q35. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q36. [1-MCQ] [1 mark]

Slip in induction motor:

- (A) Always positive
- (B) Can be negative
- (C) Always zero
- (D) Infinity

Q37. [1-MCQ] [1 mark]

DC starter used to:

- (A) Regulate speed

- (B) Limit start current
- (C) Reverse
- (D) Increase torque

Q38. [1-MCQ] [1 mark]

Slip $s =$

- (A) $(N_s - N_r)/N_s$
- (B) $(N_r - N_s)/N_s$
- (C) N_s/N_r
- (D) N_r/N_s

Q39. [1-MCQ] [1 mark]

Alternator produces:

- (A) DC
- (B) AC
- (C) Both
- (D) None

Q40. [1-MCQ] [1 mark]

Transformer works on:

- (A) Auto induction
- (B) Mutual induction
- (C) Self induction
- (D) None

Q41. [2-MCQ] [2 marks]

Max torque in IM at start ($s=1$) requires:

- (A) $R_2 = X_2$
- (B) $R_2 = 0$
- (C) $X_2 = 0$
- (D) $R_2 \gg X_2$

Q42. [2-MSQ] [2 marks]

Over-excited synchronous generator operates at:

- (A) Lagging PF
- (B) Leading PF
- (C) Unity PF
- (D) Zero PF

Q43. [2-MCQ] [2 marks]

Synchronous condenser is:

- (A) Synchronous motor at no-load
- (B) Synchronous generator
- (C) Induction motor
- (D) DC motor

Q44. [1-MCQ] [1 mark]

Synchronous generator excitation:

- (A) Rotor
- (B) Stator
- (C) Both
- (D) None

Q45. [1-NAT] [1 mark]

Most efficient transmission:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

Circuit breaker operates on:

- (A) Overload
 - (B) Short circuit
 - (C) Both
 - (D) None
-

Q47. [1-NAT] [1 mark]

Per unit system base values:

(Numerical Answer Type — Enter the numerical value)

Q48. [2-MCQ] [2 marks]

Transmission line: sending end voltage higher than receiving for:

- (A) Heavy loading
 - (B) Light loading (capacitive)
 - (C) Unity PF
 - (D) No load
-

Q49. [2-MSQ] [2 marks]

Per unit reactance: $X_{\text{ohm}}=10$, base=100 MVA, kV=132. $X_{\text{pu}} =$

- (A) 0.573
 - (B) 5.73
 - (C) 0.0573
 - (D) 57.3
-

Q50. [2-MSQ] [2 marks]

3-phase, 4-pole IM at 1440 rpm on 50 Hz. Slip:

- (A) 2%
 - (B) 4%
 - (C) 6%
 - (D) 8%
-

Q51. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \times 2$
 - (D) $P \times r/100$
-

Q53. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18

- (B) 27
- (C) 9
- (D) 3

Q54. [1-MCQ] [1 mark]

Aliasing avoided by:

- (A) Increasing rate
- (B) Decreasing rate
- (C) Both
- (D) None

Q55. [2-MCQ] [2 marks]

Nyquist rate for 4kHz signal:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 1kHz

Q56. [2-MCQ] [2 marks]

FS coefficients:

- (A) Discrete
- (B) Continuous
- (C) Both
- (D) None

Q57. [1-NAT] [1 mark]

Ideal op-amp gain:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

RC phase shift uses:

- (A) 180°
- (B) 360°
- (C) 90°
- (D) 45°

Q59. [1-NAT] [1 mark]

Negative feedback:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

ICBO in transistor is:

- (A) Collector-Base reverse
- (B) Emitter-Base
- (C) Collector-Emitter
- (D) None

Q61. [2-MCQ] [2 marks]

Displacement current added by:

- (A) Faraday
- (B) Maxwell

- (C) Ampere
- (D) Gauss

Q62. [2-MCQ] [2 marks]

Corona loss increases with:

- (A) Wire diameter
- (B) Voltage
- (C) Both
- (D) None

Q63. [1-MCQ] [1 mark]

PLD stands for:

- (A) Programmable Logic Device
- (B) Processor Logic
- (C) Programmable Logic Design
- (D) None

Q64. [1-MCQ] [1 mark]

Universal gate:

- (A) AND
- (B) OR
- (C) NAND
- (D) XOR

Q65. [1-MCQ] [1 mark]

Mod-10 counts from:

- (A) 0 to 10
 - (B) 0 to 9
 - (C) 1 to 10
 - (D) 1 to 9
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1NAT	B	1
6	1MCQ	A	1
7	1MCQ	B	1
8	1MCQ	A	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1NAT	B	1
13	1MCQ	C	1

14	1NAT	C	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1NAT	A	1
21	1MCQ	A	1
22	1MCQ	C	1
23	1NAT	A	1
24	1MCQ	B	1
25	1NAT	B	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1MCQ	C	1

29	1MCQ	B	1
30	1MCQ	A	1
31	2MCQ	B	2
32	2MCQ	A	2
33	2MCQ	B	2
34	2MCQ	C	2
35	2MCQ	B	2
36	2NAT	A	2
37	2MCQ	A	2
38	2MCQ	B	2
39	2MCQ	D	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2NAT	B	2
43	2NAT	A	2

44	2MCQ	C	2
45	2MCQ	B	2
46	2MCQ	C	2
47	2NAT	A	2
48	2MCQ	B	2
49	2MSQ	A	2
50	2MCQ	C	2
51	2NAT	A	2
52	2MCQ	A	2
53	2MSQ	B	2
54	2MCQ	C	2
55	2NAT	C	2
56	2NAT	A	2
57	2MCQ	A	2
58	2MSQ	A	2

59	2MCQ	B	2
60	2MCQ	C	2
61	2MSQ	B	2
62	2NAT	B	2
63	2MCQ	D	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: DC Machines | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Alternators | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Transformers | Difficulty: easy

Q11.

Answer: A

Solution:

Correct Answer: A

Subject: Power Systems | Topic: Generators | Difficulty: moderate

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Power Systems | Topic: Transmission | Difficulty: easy

Q13.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Protection | Difficulty: easy

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Per Unit | Difficulty: moderate

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Frequency Response | Difficulty: easy

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Root Locus | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: moderate

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Time Response | Difficulty: easy

Q19.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Choppers | Difficulty: moderate

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Inverters | Difficulty: difficult

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Converters | Difficulty: moderate

Q22.

Answer: C

Solution:

Correct Answer: C

Subject: Power Electronics | Topic: Rectifiers | Difficulty: moderate

Q23.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Op-Amps | Difficulty: moderate

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Oscillators | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Feedback | Difficulty: easy

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Transistors | Difficulty: easy

Q27.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Electronics | Topic: PLD | Difficulty: easy

Q28.

Answer: C

Solution:

Correct Answer: C

Subject: Digital Electronics | Topic: Logic Gates | Difficulty: moderate

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Electronics | Topic: Counters | Difficulty: easy

Q30.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Sampling | Difficulty: difficult

Q31.

Answer: B

Solution:

Correct Answer: B

Subject: Signals and Systems | Topic: Sampling | Difficulty: moderate

Q32.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Fourier | Difficulty: difficult

Q33.

Answer: B

Solution:

Correct Answer: B

Subject: EMFT | Topic: Maxwell | Difficulty: moderate

Q34.

Answer: C

Solution:

Correct Answer: C

Subject: EMFT | Topic: Transmission | Difficulty: easy

Q35.

Answer: B

Solution:

Type 1: $ess = 1/K_v = 1/5 = 0.2$ for unit ramp.

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q36.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: B

Solution:

Ferranti effect: light load/capacitive charging causes $V_s > V_r$. Long lines at no-load.

Subject: Power Systems | Topic: Power Systems | Difficulty: easy

Q39.

Answer: D

Solution:

Series Z: $I_1=I_2$, $V_1-V_2=Z*I_1$. $Y_{11} = I_1/V_1$ with $I_2=0$ impossible ($I_1=I_2$). $Y_{11}=0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q40.

Answer: A

Solution:

Max torque at $s = R_2/X_2$. For start ($s=1$): $R_2 = X_2$ gives max starting torque.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q41.

Answer: A

Solution:

$V_{out} = 10*5/(5+5) = 5V$. $I = 5/5 = 1A$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q42.

Answer: B

Solution:

Root at $s = -1$ (on left half plane). Simple pole on negative real axis: stable (marginally, as $t \rightarrow \infty$).

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q43.

Answer: A

Solution:

IGBT: BJT (high current) + MOSFET (easy drive). High input impedance, low saturation voltage.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q44.

Answer: C

Solution:

Root locus: starts at OL poles, ends at OL zeros (or infinity if $P > Z$).

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q45.

Answer: B

Solution:

Compare to $s^2 + 2\zeta\omega_n s + \omega_n^2$. $2\zeta\omega_n \sqrt{5} = 2$. $\zeta = 2/(2\sqrt{5}) = 1/\sqrt{5}$ approx 0.447.

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q46.

Answer: C

Solution:

Null: $R_1/R_2 = R_3/R_4$, which implies $R_1 R_4 = R_2 R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q47.

Answer: A

Solution:

Zeros from numerator: $s^2 + 4s + 3 = (s+1)(s+3)$. Zeros at $s = -1, -3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q48.

Answer: B

Solution:

$R_a = (R_1 R_2 + R_2 R_3 + R_3 R_1) / R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q49.

Answer: A

Solution:

Buck (step-down): $V_o = D V_{in}$ where D is duty cycle.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q50.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n(n-2) = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Measurements | Topic: Measurements | Difficulty: easy

Q52.

Answer: A

Solution:

Full-wave: reverse voltage across non-conducting diode = V_m .

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q53.

Answer: B

Solution:

Over-excitation: generator supplies reactive power, operates at leading power factor.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: easy

Q54.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

$h_{11} = (V_1/I_1)$ with output shorted. Units: $V/A = \text{Ohm}$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q57.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Measurements | Topic: Measurements | Difficulty: difficult

Q58.

Answer: A

Solution:

$X_{pu} = X * MVA_{base} / (kV_{base})^2 = 10 * 100 / 132^2 = 1000/17424 = 0.0574$. Hmm. With formula $X_{pu} = X * (MVA_{base} / (kV_{base}^2)) * 10: 10 * 100 / (132^2) * 10 = 10000/17424 = 0.574$. Option A.

Subject: Power Systems | Topic: Power Systems | Difficulty: difficult

Q59.

Answer: B

Solution:

$\tau = L/R = 5/10 = 0.5 \text{ s}$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q60.

Answer: C

Solution:

$N = Z - P$. $N = -1$ means one clockwise encirclement of -1 point needed. $Z = N + P = -1 + 1 = 0$. Stable.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q61.

Answer: B

Solution:

$N_{sync} = 120 * 50 / 4 = 1500$. $s = (1500 - 1440) / 1500 = 60 / 1500 = 0.04 = 4\%$.

Subject: Power Systems | Topic: Power Systems | Difficulty: difficult

Q62.

Answer: B

Solution:

Current Source Inverter: input is current source (large inductor in series).

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q63.

Answer: D

Solution:

120-degree conduction: triplen harmonics (3rd, 9th, ...) eliminated in line voltage.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: easy

Q64.

Answer: A

Solution:

Synchronous condenser: synchronous motor running at no-load, used for power factor correction.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q65.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3*3 = 18$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Control Systems: 9 questions, 14 marks
Power Electronics: 9 questions, 14 marks
Network Theory: 7 questions, 14 marks
Engineering Mathematics: 6 questions, 11 marks
Electrical Machines: 8 questions, 11 marks
Power Systems: 7 questions, 10 marks
Measurements: 3 questions, 6 marks
Signals and Systems: 3 questions, 5 marks
General Aptitude: 4 questions, 4 marks
Analog Electronics: 4 questions, 4 marks
EMFT: 2 questions, 4 marks
Digital Electronics: 3 questions, 3 marks

Difficulty Distribution:

Easy: 21 | Moderate: 35 | Difficult: 9

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.